

8 Hoare Logic

A formalism relating the initial and terminating state of a program. Written with **command** C , **precond** P and **postcond** Q . It is a deductive proof system of **triples** $\{P\} C \{Q\}$. **Assertion lang** is a lang for defining state predicates and their semantics. It is an instance of first order logic with equality, with values as integers and assertions as properties of while statements:

$$v \in \text{Vars} \triangleq \text{id} \mid \text{id}_{\log}$$

$$t \in \text{Terms} \triangleq v$$

$$P, Q \in \text{Ass} \triangleq \text{tt} \mid \text{ff} \mid P \wedge Q \mid P \vee Q \mid P \rightarrow Q \\ \mid \forall \text{id}. P \mid \exists \text{id}. P \mid t_1 = t_2$$

Note that $\neg P \triangleq P \rightarrow \text{ff}$. A term t in state s is written as $\llbracket t \rrbracket(s)$ where $\llbracket \cdot \rrbracket \subseteq \text{Terms} \times \text{State} \rightarrow \mathbb{Z}$.